PERFORMING OPERATIONS OF SETS AND SUBSETS

DIFFERENCE OF TWO SETS

- How to find the difference of two sets?
- If A and B are two sets, then their difference is given by A - B or B - A.

  - If A = {2, 3, 4} and B = {4, 5, 6}

  A - B means elements of A which are not the elements of B.

  i.e., in the above example A - B = {2, 3}

CARDINALITY

Definition:

Let S be a set. If there are exactly n distinct elements in S, where n is a nonnegative integer, we say S is a finite set and that n is the cardinality of S. The cardinality of S is denoted by | S |.

Examples:

V={1 2 3 4 5} | V | = 5

A={1,2,3,4, …, 20} |A| =20

| Ø | = 0

Solved examples to find the difference of two sets:

GIVEN:

1. A = {1, 2, 3} and B = {4, 5, 6}.

Find the difference between the two sets:

- (i) A - B
- (ii) B - A

Solved examples to find the difference of two sets:

Solution:

The two sets are disjoint as they do not have any elements in common.

(i) A - B = {1, 2, 3} = A

(ii) B - A = {4, 5, 6} = B

2. Let A = {a, b, c, d, e, f} and B = {b, d, f, g}.

Find the difference between the two sets:

- (i) A and B
- (ii) B and A

Solution:

(i) A - B = {a, c, e}

- Therefore, the elements a, c, e belong to A but not to B

(ii) B - A = {g)

- Therefore, the element g belongs to B but not A.

3. Given three sets P, Q and R such that:

P = {x : x is a natural number between 10 and 16},

Q = {y : y is a even number between 8 and 20} and

R = {7, 9, 11, 14, 18, 20}

Find the following difference of two sets

a. Find P - Q

b. Find Q – R

c. Find R - P

d. Find Q – P

Solution:

According to the given statements:

P = {11, 12, 13, 14, 15}

Q = {10, 12, 14, 16, 18}

R = {7, 9, 11, 14, 18, 20}

a. P – Q = {Those elements of set P which are not in set Q}

= {11, 13, 15}

b. Q – R = {Those elements of set Q not belonging to set R}

= {10, 12, 16}

c. R – P = {Those elements of set R which are not in set P}

= {7, 9, 18, 20}

d. Q – P = {Those elements of set Q not belonging to set P}

= {10, 16, 18}

Operations on Sets

Given sets A and B , we can define the following operations:

Operation	Notation	Meaning
Intersection	$A \cap B$	all elements which are in both A and B
Union	$A \cup B$	all elements which are in either A or B (or both)
Difference	$A - B$	all elements which are in A but not in B
Complement	$\overline{A}$ (or A^C)	all elements which are not in A

Example 1:

Let $A=\{1,2,3,4\}$ and let $B=\{3,4,5,6\}$.

Find: $A \cap B$, $A \cup B$, $A-B$, and A^C

Then:

$A \cap B=\{3,4\}$

$A \cup B=\{1,2,3,4,5,6\}$

$A-B=\{1,2\}$

$A^C=\{$all real numbers except 1,2,3 and 4$\}$

Example 2:

Let $A=\{y,z\}$ and let $B=\{x,y,z\}$.

Find: $A \cap B$, $A \cup B$, $A-B$, and A^C

Then:

$A \cap B=\{y,z\}$

$A \cup B=\{x,y,z\}$

$A-B=\emptyset$

$A^C=\{$everything except y and z$\}$

Performing Set Operations

Always perform any operations inside parenthesis first!

Given:

U = { 1, 2, 3, 4, 5, 6, 7, 8, 9, 10}

A = { 1, 3, 7, 9 }

B = { 3, 7, 8, 10 }

Find

A. (A U B)’

B. A’ ∩ B’

SOLUTION:

A.

$A \cup B = \{1, 3, 7, 8, 9, 10\}$

$(A \cup B)' = \{2, 4, 5, 6\}$

B.

$A' = \{2, 4, 5, 6, 8, 10\}$

$B' = \{1, 2, 4, 5, 6, 9\}$

$A' \cap B' = \{2, 4, 5, 6\}$